

UPINGTON, South Africa

WMO: 684240

Lat: 28.4111S

Lon: 21.2642E

Elev: 840

StdP: 91.63

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.1	1.8	-15.1	1.1	16.4	-13.2	1.3	17.6	11.6	19.3	10.2	20.3	1.4	260	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.5	39.5	17.7	38.2	17.4	37.1	17.2	21.7	29.9	21.1	29.4	20.5	29.3	4.6	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	16.2	24.0	18.9	15.2	24.1	17.9	14.2	24.4	67.8	29.7	65.4	28.7	63.0	29.3	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.6	7.5	DB	-2.5	41.7	1.2	1.0	-3.4	42.4	-4.1	43.0	-4.8	43.6	-5.7	44.3	(4)
(5)			WB	-4.5	22.9	0.9	1.4	-5.1	23.9	-5.6	24.7	-6.1	25.5	-6.8	26.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.0	28.1	27.9	25.6	21.2	17.1	13.0	12.9	14.8	18.5	22.1	24.5	26.9	(6)	
	DBStd	6.50	2.67	2.57	3.07	3.73	3.82	3.31	3.68	4.29	4.26	4.08	3.75	2.84	(7)	
	HDD10.0	34	0	0	0	0	2	9	16	7	1	0	0	0	(8)	
	HDD18.3	611	0	0	0	16	69	161	172	126	50	12	4	0	(9)	
	CDD10.0	4053	561	501	483	335	222	101	104	156	256	377	434	525	(10)	
	CDD18.3	1588	302	267	226	101	30	2	2	17	55	131	188	267	(11)	
	CDH23.3	21804	4168	3515	2728	1174	462	77	92	384	968	1876	2641	3719	(12)	
(13)	CDH26.7	11905	2506	2058	1478	504	120	3	8	108	440	978	1492	2209	(13)	
(14)	Wind	WSAvg	3.4	3.6	3.4	3.2	3.1	3.0	3.2	3.4	3.3	3.6	3.7	3.8	(14)	
(15) Precipitation	PrecAvg	197	31	37	39	23	10	4	2	2	4	11	15	17	(15)	
	PrecMax	500	159	153	134	113	47	29	27	28	23	74	82	76	(16)	
	PrecMin	53	0	0	2	0	0	0	0	0	0	0	0	0	(17)	
	PrecStd	86	36	38	33	23	12	7	5	5	6	13	16	22	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.1	40.2	39.0	35.8	31.1	27.0	27.8	31.4	36.2	38.3	39.8	40.5	(19)	
		MCWB	17.9	18.7	18.2	16.9	14.4	12.3	11.8	13.4	15.2	16.3	16.7	17.4	(20)	
	2%	DB	39.4	38.8	37.1	33.2	29.6	25.2	25.8	29.3	33.9	36.1	37.8	38.9	(21)	
		MCWB	17.9	18.2	17.6	16.5	14.0	11.7	11.4	12.7	14.5	15.8	16.4	17.2	(22)	
	5%	DB	38.0	37.2	35.8	31.9	28.0	24.0	24.1	27.8	31.5	34.3	36.0	37.2	(23)	
		MCWB	17.5	18.2	17.6	16.2	13.6	11.2	10.9	12.1	13.6	15.2	15.8	16.7	(24)	
	10%	DB	36.3	35.9	33.9	29.9	26.1	22.1	22.1	25.4	29.1	32.2	34.2	35.9	(25)	
		MCWB	17.6	18.1	17.5	16.0	13.2	10.6	10.3	11.3	12.7	14.6	15.5	16.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.7	22.7	21.9	20.3	17.1	14.0	13.5	15.2	17.4	18.8	20.4	21.1	(27)	
		MCDB	30.4	31.0	28.5	28.1	24.3	24.3	23.5	27.4	29.5	29.8	29.6	30.2	(28)	
	2%	WB	21.7	21.7	20.9	19.0	15.7	12.8	12.3	13.9	16.0	17.8	18.9	20.1	(29)	
		MCDB	29.8	30.4	29.0	26.3	24.7	23.2	23.3	26.4	28.6	29.3	30.4	30.0	(30)	
	5%	WB	21.1	21.1	20.2	18.1	14.7	11.8	11.5	12.8	14.8	16.9	17.9	19.4	(31)	
		MCDB	29.2	29.6	28.6	26.7	24.6	22.3	22.8	25.8	28.4	29.9	30.2	30.0	(32)	
	10%	WB	20.3	20.5	19.4	17.2	13.8	10.9	10.6	11.7	13.5	15.8	16.7	18.5	(33)	
		MCDB	29.5	29.4	28.2	26.5	24.1	20.9	21.4	24.1	27.1	29.8	30.4	30.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	15.5	14.9	15.1	15.1	16.1	16.6	16.7	17.9	18.0	17.0	16.7	16.4	(35)	
		MCDBR	17.8	17.0	17.3	16.7	17.6	18.6	18.6	20.0	21.1	19.5	19.5	18.3	(36)	
	5% WB	MCWBR	6.5	6.1	6.1	6.1	7.3	9.0	9.0	9.5	9.7	8.1	7.6	7.0	(37)	
		MCWBR	12.9	12.7	13.2	13.4	15.0	16.6	16.7	17.9	17.9	16.0	15.3	14.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.334	0.345	0.337	0.312	0.281	0.265	0.270	0.304	0.338	0.354	0.315	0.318	(40)	
		taud	2.475	2.450	2.455	2.502	2.570	2.584	2.550	2.413	2.302	2.301	2.475	2.495	(41)	
	Ebn at Noon		1008	979	955	933	922	918	926	926	938	959	1020	1026	(42)	
		Edh at Noon	118	118	110	95	80	74	80	102	126	135	117	116	(43)	
(44) All-Sky Solar Radiation	RadAvg		8.24	7.41	6.47	5.17	4.22	3.71	4.05	4.95	6.15	7.33	8.30	8.62	(44)	
	RadStd		0.37	0.44	0.27	0.19	0.13	0.16	0.12	0.18	0.21	0.20	0.26	0.30	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	-0.45	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	-0.45	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page